Juan Muneton Gallego

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Education

Stanford University — Palo Alto, CA
M.S. Computational Mathematics and Engineering — *Sep 2023 – Sep 2025*
Relevant Coursework: Natural Language Processing, Parallel Programming CUDA, Bayesian Statistics, Machine Learning, Numerical Linear Algebra, Design & Analysis of Algorithms, Reinforcement Learning, Convex Optimization, Stochastic Processes

Brown University — Providence, RI
B.S. Statistics (Honors) — *Sep 2017 – May 2021*
Relevant Coursework: Probability Theory, Deep Learning, Mathematical Statistics and Inference, Applied Regression Analysis, Data Structures and Algorithms, Differential Equations, Linear Programming, Statistical Learning, Probabilistic Models

Work Experience

SLAC National Acceleration Laboratory, Stanford University — Stanford, CA
Graduate Researcher — *April 2025 – Present*
- Developing transformer-based latent diffusion models to generate sparse, high-resolution (1800×1800) X-ray diffraction images using continuous representations, eliminating the need for discrete tokenization. Advised by Cong Wang and Matthias Kling.

Adobe Research — San Jose, CA
Research Engineer Intern — *Jun 2024 – Sep 2024*
- Optimized Transformer architecture with a sub-quadratic self-attention mechanism by integrating vector quantization and variational encoding, achieving up to $9\times$ decoding speedup on sequences of length 16k+ with large codebooks ($n \approx 10^4$) compared to vanilla and hyper-attention baselines. Advised by Nikos Vlassis.

Adobe Inc. — Austin, TX
Machine Learning Engineer Intern — *June 2023 – Sep 2023*
- Built a scalable data ingestion pipeline using SQL and the New Relic API to curate over 5 million performance tickets for training machine learning models on Adobe Commerce site health.
- Developed deep Bayesian neural networks and regression models to predict Apdex scores and website performance with 95% accuracy on testing datasets.
- Deployed a microservice API for serving real-time Apdex predictions to production monitoring systems.
- Engineered modules for multi-collinearity analysis, outlier detection, and neural network regularization to enhance model generalization and training stability.

R-Dex Systems Inc. — Atlanta, GA
Machine Learning Engineer — *June 2021 – June 2023*
- Implemented custom BRANF (Background Registration-Based Adaptive Noise Filtering) algorithm for radar image frame denoising, combining optical flow registration and Robust PCA to separate noise components, resulting in 15% improved neural network training accuracy and 25% faster convergence when feeding cleaned frames to object detection models, while maintaining real-time processing capabilities for operational radar systems.
- Implemented Jacobian-Based Saliency Map, Virtual Adversarial Training (VAT) and Wasserstein Adversarial Regularization Attacks for image classification models, generating adversarial perturbations during training that increased baseline accuracy by 2 percentage points while improving model robustness to input variations and adversarial attacks.
- Engineered a pipeline leveraging variational autoencoders, PCA, and ICA to detect distributional shifts in training data with 92% accuracy, enhancing dataset reliability and model robustness.

Skills

Programming: Python, C++, SQL, Java, CUDA

Technologies: Git, Docker, OpenCV, OpenMP, Pytorch, Tensorflow, Numba, Triton, scikit-learn, Linux, Spark, MLFlow, FastAPI, Wandb, Accelerate, ONNX, Nvidia Nsight, LaTeX

ML Frameworks: Supervised and Unsupervised Learning, Fine-tuning, Reinforcement Learning, RAG, Generative Modeling

Mathematical Skills: Real Analysis, Complex Analysis, Linear Algebra, Probability, Mathematical Statistics